Proceedings of the Second International Conference on Automation, Computing and Renewable Systems (ICACRS-2023)
IEEE Xplore Part Number: CFP23CB5-ART; ISBN: 979-8-3503-4023-5

2023 2nd International Conference on Automation, Computing and Renewable Systems (ICACRS) | 979-8-3503-4023-5/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICACRS58579.2023.10404790

Extraction of Emojis and Texts to Intensify Opinion Mining using Machine Learning and Deep Learning Models

Paladugu Trisha Sai
Department of Information Technology
Velagapudi Ramakrishna Siddhartha Engineering College (JNTUK)
Vijayawada, India
paladugutrishasai@gmail.com

Gudavalli Hruthi Sri
Department of Information Technology
Velagapudi Ramakrishna Siddhartha Engineering College (JNTUK)
Vijayawada, India
hruthigudavalli@gmail.com

T.Lakshmi Surekha
Department of Information Technology
Velagapudi Ramakrishna Siddhartha Engineering College (JNTUK)
Vijayawada, India
lakshmisurekha@vrsiddhartha.ac.in

***Abstract*— This research mainly focuses on product reviews by improving the customer experience. Furthermore, it allows customers to use a better-quality product according to their budget. It allows Businesses to make data-driven decisions about product development, marketing tactics, inventory management, and customer service enhancements by collecting and evaluating sentiment data. This research examines the possibility of emoji-text fusion algorithms to increase sentiment prediction accuracy considering the widespread use of emojis to represent emotion in textual material. A varied dataset of product reviews from Flipkart and Amazon, including a wide range of consumer goods and services, has been put together for this research. This research focuses on sophisticated natural language processing (NLP) methods for text analysis and emoji sentiment classification to accomplish the required goal. To properly capture both textual and emoji-based sentiment cues, This study centers on various feature extraction techniques. Additionally, this study recommends creative fusion along with machine learning and deep learning models such as logistic regression, random forest, BERT, and LSTM to find the polarity of the both texts and emojis, Hence the result is obtained with BERT model with high accuracy including texts with emojis and the outcomes are obtained using proper visualization techniques.**

Keywords*— *Emojis, texts, Sentiment Analysis, Natural Language Processing, Product reviews, Amazon and Flipkart, Polarity, Bidirectional encoders representations from Transformers (BERT), Long Short-Term Memory (LSTM), Logistic Regression, Random Forest.

I. INTRODUCTION

The beginning describes how sentiment evaluation is increased using emoji. The problem's genesis and the significance of selecting it come first. This section outlines the entire backdrop and problem definition.

A. Origin of the problem

The quantity of product reviews that are accessible online has significantly increased due to E-commerce's explosive growth [3]. Due to the growth in user-generated material, automated technologies are required to examine and draw practical conclusions from these evaluations. Product reviews are becoming a more important resource for consumers when making purchases. Because of this, companies are very interested in comprehending the opinions expressed in reviews to determine consumer happiness, spot potential improvement areas, and fine-tune their marketing tactics. It is now possible for researchers to explore sentiment analysis thanks to the accessibility of big datasets of product reviews, which are frequently publicly available on [3] websites like Amazon and Flipkart datasets. These reviews can be gathered by investigators, who can make use of them to conduct significant analysis on consumer attitudes.

The successful automation of sentiment analysis activities has been made possible by improvements in NLP methods [7] and machine learning algorithms [2]. Once a labor-intensive manual process, researchers and practitioners may now create models that can categorize the emotion expressed in text data.

Sentiment analysis has been completely transformed by machine learning and deep learning techniques, which allow computers to recognize, understand, and extract sentiment from text and emojis. Since emojis are widely used in social media posts to express feelings and opinions, these techniques have shown to be especially helpful in this area of data analysis. The trained model is fed a dataset of labelled text and emoticons after a suitable machine learning or deep learning model has been selected. In order to predict outcomes on fresh data, the model gains the ability to associate the extracted features with the appropriate sentiment labels. Metrics like accuracy, precision, recall, and F1-Score measure are used to assess the performance of the model.

The automated method of reading texts to identify the emotions (Satisfied, Neutralized or Dissatisfied,)

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

communicated is known as sentiment analysis. Customer service management, sentiment analysis, and social media monitoring are all common usage cases of sentiment analysis. Posts on social media often include some of the most honest critiques of your products, services, and businesses since they are unwanted. You can quickly go through all that data to look at the emotions of specific people as well as the general public across all social media platforms using sentiment analysis tools. Sentiment analysis can recognize sarcasm beyond the simple definition, comprehend common chatroom abbreviations (such as lol, idk, tytl etc.), and is appropriate for common syntax and spelling problems.

Two of the biggest and most well-known e-commerce platforms [3] worldwide are Amazon and Flipkart [4]. Millions of customers visit them, and they feature a huge amount of product listings and product reviews. The quantity of product reviews has considerably increased given that these platforms have grown more popular.

Opinion mining can assist firms in extracting meaningful data from customer evaluations and understanding customer preferences. Businesses may easily sift through enormous amounts of reviews and reliably discover common themes and subjects by employing brand sentiment tools.

B. Basic definition and Background

- **Sentiment Analysis** – A natural language processing (NLP) technique called sentiment analysis [7], commonly known as opinion mining [1], analyzes textual data to determine the sentiment or emotional tone portrayed within the text.
- **SpaCy Emoji** – SpaCy is a powerful program library for natural language processing that was created in the computer programming languages Python and Cython**.**
- **Emojis** – The three primary categories of satisfied, neutral, and dissatisfied comments on product reviews can be distinguished by using emoji backgrounds. Emoji's main purpose is to replace the emotional clues that text-based communication lacks.
- **Polarity** – The direction or tone of sentiment in text data is referred to as polarity, which can be categorized as positive, negative, or neutral
- **BERT Model** – The subject of text analysis and comprehension has been completely transformed by the innovative natural language processing (NLP) approach known as BERT [8] (Bidirectional Encoder Representations from Transformers).
- **Evaluation Measures** –There are several assessment metrics some of them include Accuracy, Precision, Recall, F1 Score, Confusion Matrix etc.
- **LSTM** – A specific form of recurrent neural network (RNN) architecture known as Long Short-Term Memory (LSTM) [8] has excelled at several natural language processing (NLP) tasks [7], including sentiment analysis, which involves classifying communications.
- **Logistic Regression** – A categorize data into two or more groups is logistic regression [10]. A labeled dataset of product reviews, where each comment is given one of the three sentiment categories, is commonly used to train a logistic regression model for this job. During training, the model learns to associate the text data's attributes with the appropriate sentiment category.
- **Random Forest** – Powerful machine learning [2] algorithm Random Forest [10] is used to group comments on product reviews into various emotion categories. Essentially, Random Forest starts by constructing a bunch of decision trees, each trained on a portion of the dataset and utilizing a randomly chosen set of features. Each of these decision trees offers its predictions for the tone of the product reviews. The program combines these predictions using a voting method to reach a conclusion.
- **NLP** – The application of artificial intelligence to the interplay of computers and human language is known as natural language processing, or NLP [7].

C. Problem Statement, Objectives, and Outcomes

Understanding public sentiment toward diverse topics, goods, or services requires performing opinion mining or sentiment analysis [1]. But a lot of opinion mining systems only concentrate on text data, frequently ignoring the rich information provided by emojis. Emojis are becoming an essential component of modern communication and have a significant effect on how text is expressed emotionally.

The objective of the study includes to Collect reviews from the large E-Commerce platforms [3] such as Amazon, Flipkart, and fusion them to extract emojis and texts to obtain polarity of the reviews and categorize them into satisfied, unsatisfied, and neutral sentiments.

D. Real-time Applications of Proposed Work

This technique can give real-time perceptions of people's attitudes regarding particular subjects, brands, or events, for example, through social media monitoring. It can recognize popular emojis and the written comments that go along with them, assisting marketers and analysts

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

in quickly and accurately gauging public sentiment. Real-time marketing tactics and crisis management initiatives can be guided by this knowledge. Businesses can better understand client emotions and concerns by extracting and interpreting emojis along with textual comments in customer care and feedback analysis. This may result in faster response times and more client satisfaction. This method can be used in political analysis and online forums to monitor voter opinion during elections, political debates, or policy discussions. Analysts can discover more about the emotional tone being transmitted through text and emoticons, which can be extremely useful for campaign strategy and policies.

Opinion mining [1] initiates by tokenizing text data, breaking it into smaller units like words, phrases, or sentences. This segmentation aids in understanding the fundamental components of language, enabling the analysis of sentence structure and word meanings. Sentiment analysis simplifies the examination of emotions, categorizing them as neutral, negative (Dissatisfied), or positive (Satisfied), assigning numerical scores for various business applications, such as marketing and brand monitoring. Named entity recognition focuses on identifying terms in text, like people, places, and company names. Feature extraction methods include TF-IDF, which prioritizes words that are frequent within a document but rare in the entire corpus, enhancing their informativeness.

II. RELATED WORK

According to this research, emojis are widely used on social media platforms to express emotions. The study also suggests that new emojis are being developed to express mixed emotions. Opinion mining [1] is becoming increasingly important in extracting knowledge from social media data to support content decisions1. The study reviewed indicate that emotions expressed through emojis can be translated and used for accurate decision-making.

A. *The Essential of Sentiment Analysis and Opinion Mining in social media: Introduction and Survey of the Recent Approaches and Techniques.*

Due to the extensive use of social media platforms such as Facebook, Twitter, and YouTube, sentiment analysis and opinion mining have grown rapidly. It is a branch of natural language processing and data mining, particularly web mining and text mining.

B. *Evaluation of Deep Learning Techniques in Sentiment Analysis from Twitter Data*

This study compares several deep learning techniques used for sentiment analysis in Twitter data. Neural networks, which are effective in image processing, recurrent neural networks (RNN), which are used in natural language processing (NLP) tasks, and deep learning (DL) techniques, which contribute to the solution of a wide range of problems, have gained popularity in this field.

C. *A Deep Learning Model Enhanced with Emojis for Sina-Microblog Sentiment Analysis*

According to a research study, emojis are important in Sina Microblogs as they can convey meaning and emotions effectively in environments with character limitations. Emojis can add context and nuance to text messages, thereby enhancing the sentiment. The study proposes a deep learning approach that uses emojis along with textual data for sentiment analysis to address this issue.

D. *Opinion mining for emotions determination*

This study aims to understand how emotions are conveyed in various types of user-generated content, such as online comments, social media posts, and product reviews. The study examines the challenges of detecting emotions in the context of opinion mining [1].

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

III. PROPOSED WORK

A. *Architecture Diagram*

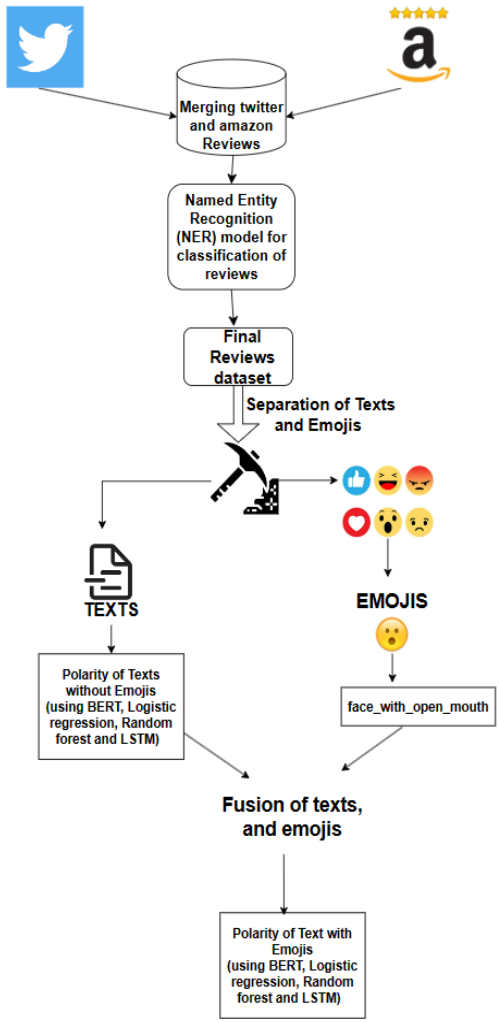

Fig. 1. *Architecture Diagram*

Brief explanation of Architecture Diagram:

1. Collect the customer reviews from the Amazon and Flipkart datasets.
2. The reviews are then stored in a excel sheet and that sheet is used as Dataset.
3. Using the Named Entity Recognition model, the final dataset have been filtered that contains reviews only about Air Pods second and third generation.
4. Separate the emojis using regular expressions or the designated Unicode character ranges, extract the emojis from the text.
5. Text is drawn out from the reviews using Spacy or NLTK library.
6. Preprocessing of texts is performed using normalization Discretization, tokenizing, root form, remove stop words.
7. Polarity of the text without emojis is determined by using Text Blob library.
8. Polarity of the texts with emojis is calculated by using Text Blob library.
9. Based on the polarity score the feelings are categorized as, if polarity is less than zero then it is disssatisfied, if equal to zero then Neutralized else it is Ssatisfied.
10. After classifying the sentiments, the performance metrics evaluation is calculated using Logistic Regression, Random Forest, BERT, and LSTM models.

B. *Description of the algorithm*

Long Short-Term Memory (LSTM):

A common recurrent neural network (RNN) architecture used in natural language processing (NLP) tasks, like sentiment analysis on product evaluations, is called Long Short-Term Memory (LSTM).
Input: Product Reviews from Amazon and Flipkart Datasets.
Output: The results could include labels with the words "satisfied," "neutralized," or "dissatisfied" in them.

Input Gate (i_t): An LSTM cell has several parts at Each time step t, like an output gate, cell state, forget gate, and input gate. These components are computed using the following formulas:

$$i_t = \text{sigmoid}\ (W_i * [h_(t-1), x_t] + b_i)$$

Where:

- i_t is input gate activation.
- [h_(t-1), x_t] represents the concatenation of the previous hidden state (h_(t-1)) and the current input (x_t).
- W_i and b_i are weights and biases for the input gate.

Forget Gate (f_t):

$$f_t = \text{sigmoid}\ (W_f * [h_(t-1), x_t] + b_f)$$

Where:

- f_t is the forget gate activation.
- W_f and b_f are learnable weights and biases for the forget gate.

Cell State Update or candidate memory (C~_t):

$$C\sim t = \tanh\ (W_c * [h(t-1), x_t] + b_c)$$

Where:

- C~_t is the candidate cell state.
- tanh is the hyperbolic tangent function.
- W_c and b_c are learnable weights and biases for updating the cell state.

Output Gate (o_t):

$$o_t = \text{sigmoid}\ (W_o * [h_(t-1), x_t] + b_o)$$

Where:

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

- o_t is the output gate activation.
  W_o and b_o are learnable weights and biases for the output gate.

Hidden State (h_t):

$$h_t = o_t * \tanh(C_t)$$

Where:

- h_t is hidden state at time step t.

These equations describe the sequential flow of data through an LSTM cell. The information that is sent to the hidden layer of the following time step is decided by output gate, how long previous information is retained is controlled by the forget gate, and input gate controls the flow of new information.

BERT: A pre-trained transformer-based model called the BERT (Bidirectional Encoder Representations from Transformers) technique was created for a variety of natural language processing (NLP) problems.

Formula for a transformer encoder:

On top of the Transformer architecture, BERT is built.

The following equation is useful for figuring out a Transformer encoder layer's output:

$$\text{Attention}(Q, K, V) = \text{SoftMax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where:

- Q: Query Matrix
- K: Key Matrix
- V: Value Matrix
- d_k: Dimension of key vectors.

Multi-Head Self-Attention:

To capture various facets of the input text, BERT employs multi-head self-attention. For every attention head, there is a distinct set of parameters that can be learned. The multi-head self-attention equation is as follows:

$$\text{MultiHead}(Q, K, V) = \text{Contact}(\text{head}_1, \ldots\ldots \text{head}_h)W^o$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Where:

- h: Number of attention heads
- W_i^Q, KW_i^K, VW_i^V : Learnable weight matrices for each head
- W^o : Learnable weight matrix for the output.

Contextual understanding, tokenization, text preprocessing, and fine tuning are all used by the BERT model. Its job is to obliterate stop words, extraneous noise, and punctuation—emojis and not. The next step in the tokenization process is treating emojis like tokens. BERT converts each token—word, subword, or emoji—into high-dimensional vector representations that express the word context of the text. Being a bidirectional model, it can read text both from left to right and from right to left. This bidirectionality helps BERT properly grasp the context of words. After pretraining on a large corpus of text, BERT is tuned on a specific sentiment analysis task using labeled data. The contextual embeddings are correlated with sentiment labels (satisfied, neutral, and dissatisfied) by the model. Masked Language Model (MLM) and Next Sentence Prediction (NSP) are the two key tasks involved in the BERT-specific pre-training.

While the NSP objective determines whether two sentences are sequential, the MLM objective seeks to predict masked words in a sentence. Cross-entropy loss for predicting masked words based on MLM Loss Formula calculations. The NSP Loss Formula uses cross-entropy loss to determine if two sentences are sequential or not.

The formula for MLM loss is like cross-entropy loss:

$$\text{MLM Loss} = -\Sigma(\log(P(x_i \mid \delta)))$$

$P(x_i \mid \delta)$ represents the expected probability of the correct token x_i, while δ denotes the parameters of the model.

C. Description of Datasets, Requirements and Tools

Social Platform Reviews - A dataset of 1000 rows are collected by considering Apple Air Pods product reviews.

Reviews
Got the genuine product, assembled in Vietnam instead of China for change. I am used to Senn
Reviewing after 4 months of usage. The airpods pro gen 2 are absolutely awesome. It sounds a
Being an AppleCompany product user, I was very happy to find this was in genuine product and
Great accessory from Apple Organization. And good delivery from Flipkart. Got this for amazin
I have been using AirPods 2 from past 1 year and I have to say when it?? a noticeable upgrade!
Awesome ????
His touch is not working and volume is come outside and not working good
Love the product, I was already using Airpods Pro for the last 4 year, this feels almost good as p
But one problem with this product as i got the only 10 months warranty as i purchased this in n
Amazing sound quality.
Good battery life.

Fig. 2. *A sample of reviews datasets*

C. Requirements

Google Collaboratory - Google Collaboratory, is a free cloud-based tool that Google offers for creating and executing Python-based machine learning and data science experiments.

Python Libraries:

- Keras – Python based Keras is an open-source framework for deep learning [6]. It may be used by both inexperienced and seasoned machine learning practitioners [14].
- NLP – Related text preparation tools are available through NLTK.
- TextBlob - This program makes routine text processing operations easier. Text preparation is where Text Blob shines.

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

- Matplotlib - Python's Matplotlib package is the industry standard for data visualization. Line plots, scatter plots, bar plots, histograms and pie charts.
- NumPy – The Python library NumPy, which stands for "Numerical Python," is the foundational library for numerical and mathematical operations.
- Demoji - Emojis and emoticons are extracted from text data using the Python module Demoji.

IV. RESULT ANALYSIS

A. Results

This chapter shows different visualization techniques that are taken to estimate different types of Models. The central theme of this study focuses on the Heatmaps and Bar graphs to visualize the comparison between the different models and compare the polarities between the texts and emojis.

```
text1=df["Reviews"][4]
text2= NER(text1)
print(text2)
for word in text2.ents:
  print(word.text,"-->",word.label_)
```

```
I have been using AirPods 2 from past 1 year and I have to say when it?
AirPods 2 --> PRODUCT
past 1 year --> DATE
the AirPods Pro --> PRODUCT
The Active Noise Cancellation --> LAW
```

Fig. 3. *Named entity recognition for one of the reviews*

From figure 3 using the Named Entity Recognition the reviews have been filtered that contains the product value as Air pods Second and third generation. The product-specific and temporal sentiment data from reviews using the output from named entity recognition are extracted and then classified this NER model to preprocess the step more effectively and used this output as the next input for the futher sentiment analysis.

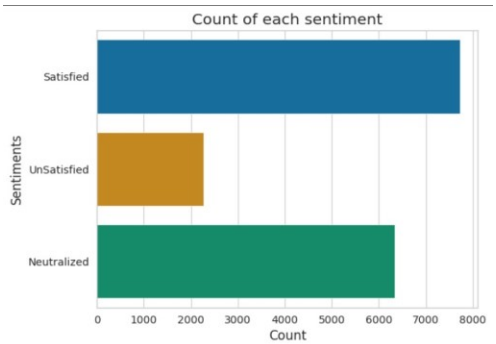

Fig. 4. *Bar graph representation of Count of each sentiment in text along with emojis*

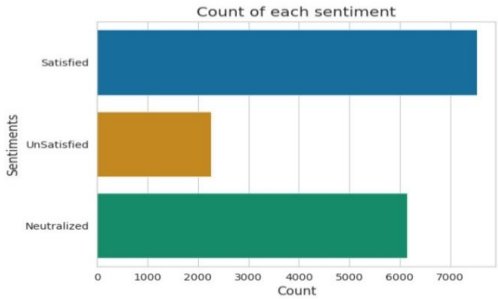

Fig 5. *Bar graph representation of Count of each sentiment in text without emojis.*

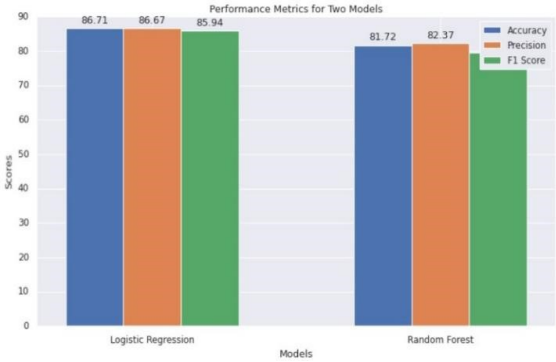

Fig. 6. *Performance Metrics Depiction of Regression and Random Forest of text without emojis*

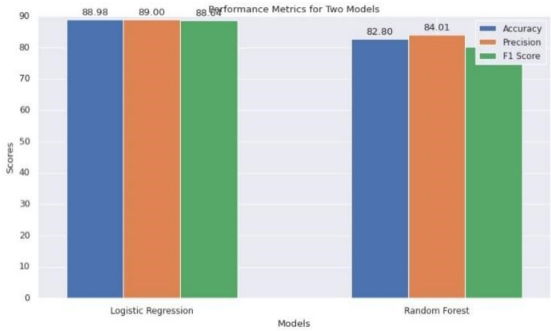

Fig. 7. *Performance Metrics Depiction of Regression and Random Forest of text with emojis*

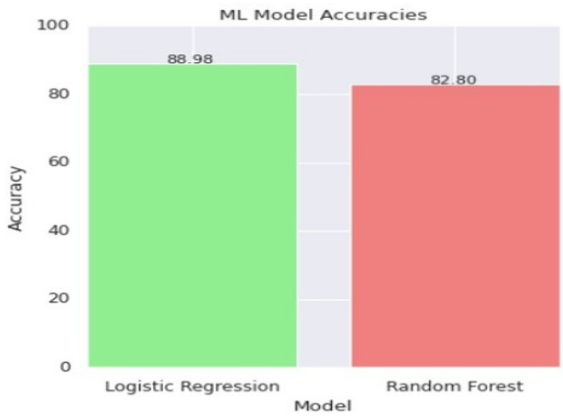

Fig. 8. *ML Model Accuracies of text with emojis*

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

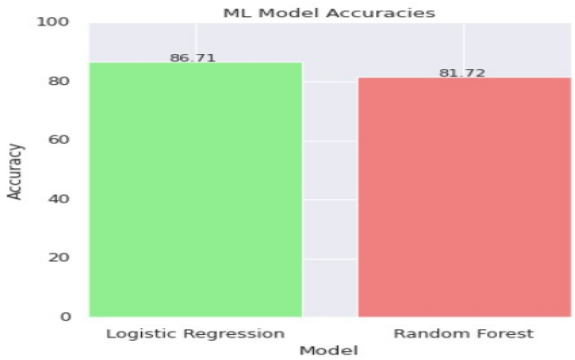

Fig. 9. *ML model accuracies of text without emojis*

```
r = model.fit(pad_train, y_train, validation_data=(pad_test, y_test), epochs=10)

Epoch 1/10
358/358 [==============================] - 349s 957ms/step - loss: 0.1666 - accuracy: 0.5676 - val_loss: -1.1418 - val_accuracy: 0.7309
Epoch 2/10
358/358 [==============================] - 363s 1s/step - loss: -6.6507 - accuracy: 0.7766 - val_loss: -15.6139 - val_accuracy: 0.7386
Epoch 3/10
358/358 [==============================] - 354s 990ms/step - loss: -30.1290 - accuracy: 0.7804 - val_loss: -44.7586 - val_accuracy: 0.7447
Epoch 4/10
358/358 [==============================] - 374s 1s/step - loss: -70.0285 - accuracy: 0.8041 - val_loss: -87.7065 - val_accuracy: 0.7748
Epoch 5/10
358/358 [==============================] - 367s 1s/step - loss: -124.5534 - accuracy: 0.8088 - val_loss: -139.6290 - val_accuracy: 0.7529
Epoch 6/10
358/358 [==============================] - 360s 1s/step - loss: -187.5980 - accuracy: 0.7851 - val_loss: -179.6109 - val_accuracy: 0.7476
Epoch 7/10
358/358 [==============================] - 359s 1s/step - loss: -249.5110 - accuracy: 0.8033 - val_loss: -251.4445 - val_accuracy: 0.7850
Epoch 8/10
358/358 [==============================] - 336s 936ms/step - loss: -332.6018 - accuracy: 0.8219 - val_loss: -316.3393 - val_accuracy: 0.7735
Epoch 9/10
358/358 [==============================] - 354s 988ms/step - loss: -417.0230 - accuracy: 0.8160 - val_loss: -388.1356 - val_accuracy: 0.7621
Epoch 10/10
358/358 [==============================] - 354s 991ms/step - loss: -518.4904 - accuracy: 0.8269 - val_loss: -489.0392 - val_accuracy: 0.7770
```

Fig. 10. *LSTM model's accuracy of 82.6% for text without emojis*

```
# training the model
r = model.fit(pad_train, y_train, validation_data=(pad_test, y_test), epochs=10)

Epoch 1/10
349/349 [==============================] - 347s 985ms/step - loss: 0.1815 - accuracy: 0.5510 - val_loss: -0.9590 - val_accuracy: 0.7518
Epoch 2/10
349/349 [==============================] - 333s 955ms/step - loss: -8.1280 - accuracy: 0.7662 - val_loss: -18.2347 - val_accuracy: 0.7884
Epoch 3/10
349/349 [==============================] - 334s 956ms/step - loss: -41.3462 - accuracy: 0.7691 - val_loss: -56.9310 - val_accuracy: 0.7931
Epoch 4/10
349/349 [==============================] - 333s 956ms/step - loss: -93.6885 - accuracy: 0.7924 - val_loss: -112.6712 - val_accuracy: 0.8083
Epoch 5/10
349/349 [==============================] - 335s 960ms/step - loss: -160.3981 - accuracy: 0.7990 - val_loss: -167.2049 - val_accuracy: 0.7732
Epoch 6/10
349/349 [==============================] - 338s 966ms/step - loss: -243.5311 - accuracy: 0.8169 - val_loss: -250.3783 - val_accuracy: 0.7972
Epoch 7/10
349/349 [==============================] - 360s 1s/step - loss: -339.0154 - accuracy: 0.8226 - val_loss: -307.1282 - val_accuracy: 0.7253
Epoch 8/10
349/349 [==============================] - 390s 1s/step - loss: -441.2424 - accuracy: 0.8236 - val_loss: -425.4883 - val_accuracy: 0.8002
Epoch 9/10
349/349 [==============================] - 374s 1s/step - loss: -561.0273 - accuracy: 0.8328 - val_loss: -529.1653 - val_accuracy: 0.8113
Epoch 10/10
349/349 [==============================] - 395s 1s/step - loss: -686.5320 - accuracy: 0.8360 - val_loss: -635.1063 - val_accuracy: 0.8175
```

Fig. 11. *LSTM model's accuracy of 83.6% for text with emojis*

TABLE I. *COMPARISON OF THE ACCURACY BETWEEN THE DIFFERENT MODELS*

Models	**Accuracy %**	
	Text without Emoji's	**Text with emoji's**
Logistic Regression	86.7	88.9
Random Forest	81.7	82.7
LSTM	82.6	83.7
BERT	95.3	96.2

```
metrics = metric)

[38] history = model.fit(
       x ={'input_ids':x_train['input_ids'],'attention_mask':x_train['attention_mask']} ,y = y_train,epochs=3,batch_size=36)

     Epoch 1/3
     28/28 [==============================] - 14s 512ms/step - loss: 0.5303 - balanced_accuracy: 0.8200
     Epoch 2/3
     28/28 [==============================] - 15s 520ms/step - loss: 0.3076 - balanced_accuracy: 0.9120
     Epoch 3/3
     28/28 [==============================] - 15s 543ms/step - loss: 0.1870 - balanced_accuracy: 0.9560
```

Fig. 12. *BERT model's accuracy of 95.3% for text without emojis*

```
Epoch 1/10
1099/1099 [==============================] - 319s 289ms/step - loss: 1.3818 - accuracy: 0.6028
Epoch 2/10
1099/1099 [==============================] - 314s 286ms/step - loss: 0.5808 - accuracy: 0.8217
Epoch 3/10
1099/1099 [==============================] - 311s 283ms/step - loss: 0.3913 - accuracy: 0.8766
Epoch 4/10
1099/1099 [==============================] - 312s 284ms/step - loss: 0.2947 - accuracy: 0.9061
Epoch 5/10
1099/1099 [==============================] - 310s 282ms/step - loss: 0.2383 - accuracy: 0.9236
Epoch 6/10
1099/1099 [==============================] - 309s 281ms/step - loss: 0.1967 - accuracy: 0.9357
Epoch 7/10
1099/1099 [==============================] - 311s 283ms/step - loss: 0.1692 - accuracy: 0.9443
Epoch 8/10
1099/1099 [==============================] - 310s 282ms/step - loss: 0.1471 - accuracy: 0.9505
Epoch 9/10
1099/1099 [==============================] - 310s 282ms/step - loss: 0.1299 - accuracy: 0.9569
Epoch 10/10
1099/1099 [==============================] - 314s 286ms/step - loss: 0.1102 - accuracy: 0.9629
```

Fig. 13. *BERT model's accuracy of 96.2% for text with emojis*

B. *Observations*

The core of this research is to classify product review comments as satisfied, neutralized, or dissatisfied by considering both text and emojis. It also evaluated the effectiveness of various models, including BERT (Bidirectional Encoder Representations from Transformers), LSTM (Long Short-Term Memory), Random Forest and logistic regression using regular expressions (logistic regex) with reference to sentiment analysis. These are the following observations that are emerged from the reviews.

With emojis, it reaches the highest accuracy of 96.2%. It is foreseeable that using BERT model with emojis can ensure that the classification of texts with emojis is high accurate. Modern transformer-based natural language understanding model BERT is well renowned for achieving remarkable results in a variety of NLP tasks [7]. This model exhibits the highest accuracy, demonstrating its efficiency in capturing the subtleties of mood in product reviews. Emojis are frequently used to express emotions, therefore this model's strong performance in both instances (with and without emojis) implies that it can handle a numerous user-generated content types, making it appropriate for tasks where this is the case.

However, while deploying BERT, it is valuable to consider issues like computational resources and model complexity because it might be more recourse-intensive than conventional models like logistic regression or random

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

forests. Additionally, the model opted accordingly should fit the demands and limitations of this study. BERT is an excellent choice for accurate sentiment analysis for product reviews.

V. CONCLUSION AND FUTURE STUDIES

A. Conclusion

This research focuses on forming extraction of emojis and text from Flipkart and Amazon reviews can enhance opinion mining by providing additional contextual information about the sentiment and emotions expressed in the reviews. By using Named Entity Recognition, the reviews that contains the product information only about Air pods second generation and third generation are filtered. After this feature extraction the Deep learning and Machine learning techniques, such as BERT, LSTM , Logistic regression and Random Forest respectively , can be used to extract features from both the text and emojis. Therefore, in this study the BERT transformer and LSTM Model are used to classify the sentiment of the review.

Overall, incorporating emojis into opinion mining using deep learning techniques can improve the accuracy and granularity of sentiment analysis, as emojis often convey emotions and sentiments that are difficult to express through text alone. BERT model has given more Accuracy when compared to other models. So, the BERT model is preferred. It is crucial to remember that the specific application and the quality of the data will determine how effective this method is, and further research is needed to fully understand the potential benefits and limitations of incorporating emojis into opinion mining.

B. Further Studies

In the future, the extraction of emojis and text from Twitter and Amazon reviews using deep learning techniques is promising. As more and more data become available, the accuracy of sentiment analysis algorithms can be expected to improve. Here are some possible future scopes for this area of research, they are Multimodal analysis, Real-time sentiment analysis, Personalized sentiment analysis, Emotion Detection, Applications beyond Opinion mining.

Furthermore, this study would also like to encourage the same research with Stickers and Gifs as they are demanding and rapidly increasing in social media. Stickers and GIFs frequently convey feelings more clearly than text alone since they are filled with emotional substance. The development of models that can recognize and interpret the emotions conveyed by these visual components may be necessary for the future of sentiment analysis. Future studies might concentrate on building multimodal sentiment analysis algorithms that combine data from several sources to increase accuracy.

To enhance the effectiveness of user perspective acquisition across diverse languages and cultures, several key strategies can be employed. Firstly, addressing emoji challenges involves translating emojis into textual descriptions across languages and employing emotion detection techniques to understand the sentiments conveyed. Secondly, it is vital to curate multilingual and multicultural datasets, incorporating samples from various regions and languages to train the model effectively. Lastly, text extraction and preprocessing steps, such as language identification and character encoding, ensure the system can handle different linguistic contexts, minimizing issues and fostering a more culturally sensitive opinion mining process.

REFERENCES

[1] Sancheng Peng a, Lihong Caob, Yongmei Zhou c, Zhouhao Ouyang d, Aimin Yang e, Xinguang Li a, Weijia Jia f, Shui Yugv "A survey on deep learning for textual emotion analysis in social networks" published on 8 October 2022

[2] Rajnish Pandey & Jyoti Prakash Singh "BERT-LSTM model for sarcasm detection in code-mixed social media post" published on October 2022

[3] Mohammad Eid Alzahrani, 1 Theyazn H. H. Aldhyani, 2 Saleh Nagi Alsubari, 3 Maha M.Althobaiti, "Developing an Intelligent System with Deep Learning Algorithms for Sentiment Analysis of E-Commerce Product Reviews" Published on 28 May 2022.

[4] Mohammad Aman Ullah, Syeda Maliha Marium, ShamimAra Begum, Nibadita Saha Dipa "An algorithm and method for sentiment analysis using the text and emoticon" published on December 2020.

[5] Leeja Mathew; V R Bindu"A Review of Natural Language Processing Techniques for Sentiment Analysis using Pre-trained Models" published on March 2020.

[6] Kanish Shah, Henil Patel, Devanshi Sanghvi & Manan Shah "A Review of Natural Language Processing Techniques for Sentiment Analysis using Pre-trained Models" published on December 2020.

[7] Pankaj. Prashant Pandey, Muskan, Nitasha Soni "A Comparative Analysis of Logistic Regression, Random Forest and KNN Models for the Text Classification" published on February 2019.

[8] Rajkumar S. Jagdale, Vishal S.Shirsat & Sachin N. Deshmukh "Sentiment Analysis on Product Reviews Using Machine Learning Techniques" published on 12 August 2018.

[9] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, "Electron spectroscopy studies on magneto-optical media and plastic substrate interface," IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].

[10] Gabriele de Seta, "The circulation of emoticons, emoji, stickers, and custom images on Chinese digital media stickers, and custom images on Chinese digital media platforms" published on 3 September 2018.

[11] Zeenia Singla; Sukhchandan Randhawa; Sushma Jain "Sentiment analysis of customer product reviews using machine learning" published in: 2017

[12] Manalee Datar & Pranali Kosamkar, "A Novel Approach for Polarity Determination Using Emoticons: Emoticon-Graph" published on 26 February 2016.

[13] Anuj Sharma, Shubhamoy Dey "Performance Investigation of Feature Selection Methods and Sentiment Lexicons for Sentiment Analysis" announced on September 2013

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.

[14] Francesco Mureddu, David Osimo, "Research Challenge on Opinion Mining and Sentiment Analysis" publication September 2012

[15] Anuj Sharma, Shubhamoy Dey, "A comparative study of feature selection and machine learning techniques for sentiment analysis" published on 23 October 2012.

Authorized licensed use limited to: University of Auckland. Downloaded on April 12,2026 at 00:49:31 UTC from IEEE Xplore. Restrictions apply.